

QUIC: Quick UDP Internet Connections

Introduction

Overview

- QUIC is an **application-layer protocol** built on **UDP**.
- Developed by Google to improve HTTP performance and reduce latency.
- Widely deployed in Google services, Chrome, and YouTube mobile apps.

Motivation

Why QUIC?

Traditional **TCP + TLS** requires:

- TCP handshake: establishes reliability and congestion control.
- TLS handshake: establishes authentication and encryption.
- Total **2 RTTs** for full connection establishment.

QUIC combines reliability, congestion control, authentication, and encryption into **1 handshake (1 RTT)**, with **0-RTT** support for repeat sessions.

Key Features

- 1. Single-RTT Handshake:** Establishes reliability, congestion control, authentication, and encryption in **1 RTT**.
- 2. 0-RTT Handshake:** Reuse previous session tickets to eliminate handshake delay for repeated connections.
- 3. Built-in Reliability and Congestion Control:** Uses algorithms similar to TCP to detect losses and control congestion.
- 4. Integrated Security:** Authentication and encryption built directly into the protocol.

QUIC vs TCP + TLS

Feature	TCP + TLS	QUIC
Reliability	TCP ensures delivery	QUIC provides on UDP
Congestion Control	TCP algorithms	QUIC algorithms
Authentication	TLS handshake	Built-in
Encryption	TLS handshake	Built-in
Handshake RTT	2 RTTs	1 RTT (0-RTT possible)

Connection Establishment Diagram

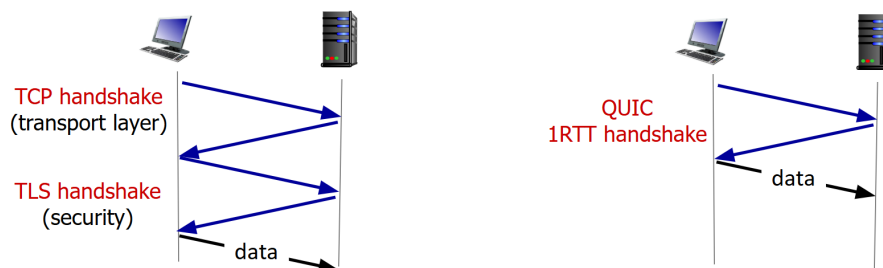


Figure 1: TCP vs QUIC: Handshake and Connection Establishment Comparison

Summary

- QUIC improves HTTP performance by reducing handshake latency and combining TCP + TLS functionality.
- Enables **1-RTT** and **0-RTT** connection establishment.
- Integrates reliability, congestion control, authentication, and encryption.
- Widely adopted in modern web services and applications.